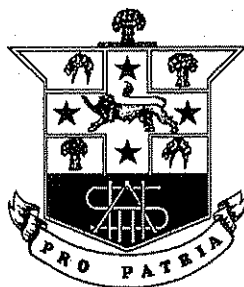


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS

EXTENSION 1

2011

YEAR 11

YEARLY (TASK 3) EXAMINATION

EXAMINERS ~ S. FAULDS, J. DILLON, P. BICZO, D. CRANCHER, G. RAWSON,
S. HACKETT

GENERAL INSTRUCTIONS

- | | |
|--|---|
| <ul style="list-style-type: none">• Reading Time – 5 minutes.• Working Time – 1 hour 30 minutes.• Attempt all questions.• Each question is worth 10 marks.• All necessary working should be shown in every question.• This paper contains eight (8) questions.• Board approved calculators and MathAids may be used. | <ul style="list-style-type: none">• Each question is to be started in a new answer booklet. Write the question number and your student number at the top of each answer booklet.• You must hand in an answer booklet for each question even if a question has not been attempted.• This examination must NOT be removed from the examination room |
|--|---|

STUDENT NAME: _____

TEACHER: _____

QUESTION 1 **10 marks** Start a NEW answer booklet

Marks

- (a) Evaluate $\log_2 5$ correct to four decimal places. 2
- (b) Find $3 \times 2^{2.5}$ correct to three significant figures. 1
- (c) Express with a rational denominator $\frac{1+\sqrt{3}}{2-\sqrt{3}}$. 2
- (d) Solve the inequality $\frac{3-2x}{x-3} < 4$. 3
- (e) Prove that $a + \frac{1}{a} \geq 2$ when $a > 0$. 2

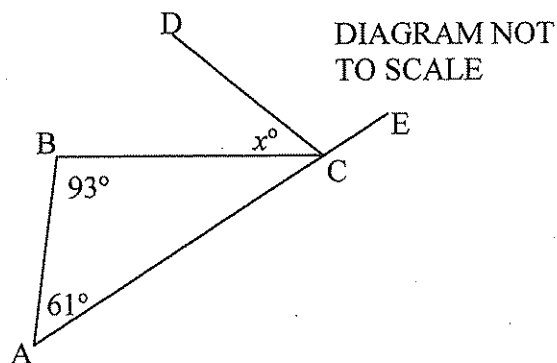
QUESTION 2

10 marks

Start a NEW answer booklet

Marks

(a)



2

Find $\angle BCD$ in the diagram above, given DC bisects $\angle BCE$. Justify your answer by showing all reasoning.

- (b) ABC is a triangle, right-angled at B. X is a point on BC such that $AX=XC$ and $\angle BAX = 2\angle XAC$.

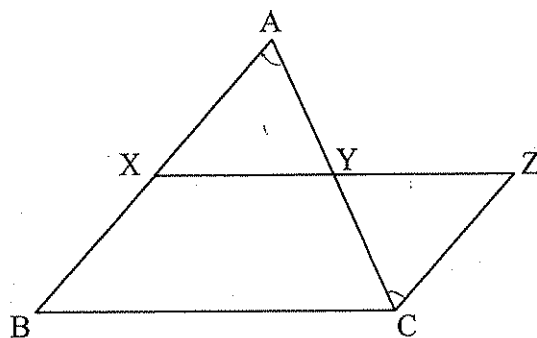
(i) Draw a neat sketch of $\triangle ABC$, showing all of the given information.

1

(ii) Find the size of $\angle ACB$, giving reasons.

2

- (c) The diagram below shows parallelogram BXZC with BX produced to A. Y is the midpoint of AC.



(i) Prove $\triangle AXY \equiv \triangle CZY$.

3

(ii) Hence, show that X is the midpoint of AB.

2

QUESTION 3

10 marks

Start a NEW answer booklet

Marks

- (a) Without using a calculator, show that:

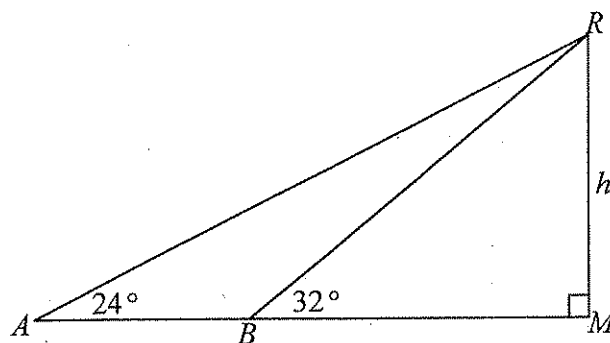
2

$$\frac{2 \tan \frac{\pi}{6}}{1 - \tan^2 \frac{\pi}{6}} = \tan \frac{\pi}{3}$$

- (b) Show that $\sin 75^\circ = \frac{1 + \sqrt{3}}{2\sqrt{2}}$.

2

- (c) A walker walks along the ground directly towards a distant high rocky cliff R . At a point A the angle of elevation of the cliff is 24° . At point B , a kilometre closer to the cliff, the angle of elevation is 32° .



- (i) Find the horizontal distance from B to the cliff, to the nearest metre.
- (ii) Find the height of the cliff above ground level, to the nearest metre.

2

1

- (d) Prove that $\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta$.

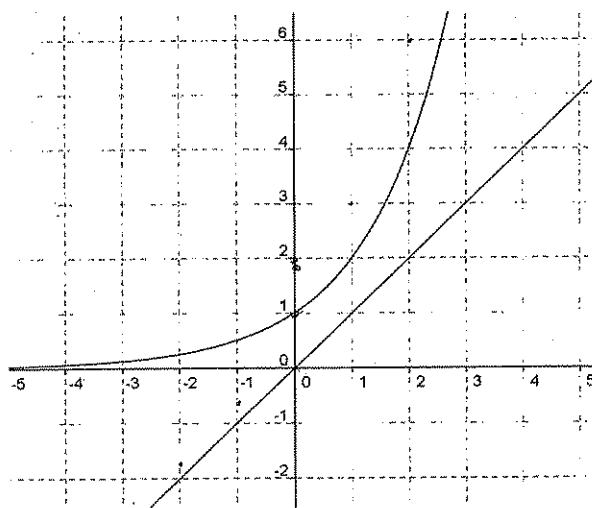
3

QUESTION 4**10 marks**

Start a NEW answer booklet

Marks

- (a) (i) State the definition of an odd function. 1
- (ii) Show that $f(x) = 2x^3$ is an odd function. 1
- (b) For the function $y = x^2 + 4x$:
- (i) Find the equation of the axis of symmetry of the parabola. 1
- (ii) Find the co-ordinates of the vertex of the parabola and hence, state the range of the function. 2
- (c) (i) Sketch the graph of $y = |x + 2|$, showing all important features. 2
- (ii) Hence, solve $|x + 2| > 4 - |x|$. 2
- (d) Copy or trace the following curves onto your answer sheet. 1

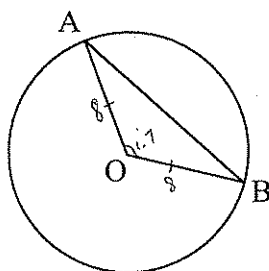
Using addition of ordinates, sketch the curve $y = 2^x + x$.

QUESTION 5**10 marks**

Start a NEW answer booklet

Marks

- (a) (i) Convert 36° to radians, expressing your answer in terms of π . 1
- (ii) In the diagram below, the circle has a radius of 8cm and $\angle AOB = 1.7$ radians. 2



Find the area of the minor segment enclosed by the chord AB.

- (b) Differentiate the following with respect to x :

(i) $5x^4 - \frac{4}{x}$

2

(ii) $\sqrt{2x+7}$

2

- (c) Prove, by the process of mathematical induction, that:

3

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

for all positive integers, n .

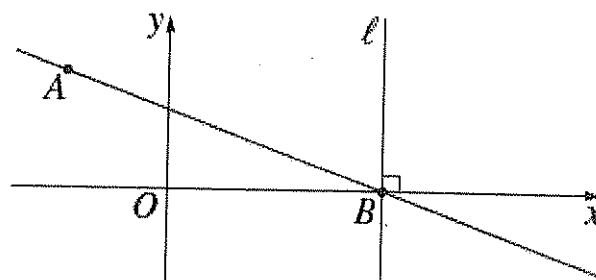
QUESTION 6

10 marks

Start a NEW answer booklet

Marks

- a) The diagram shows the points $A(-1,3)$ and $B(2,0)$.
The line l is drawn perpendicular to the x -axis through the point B .



- | | |
|--|---|
| (i) Calculate the length of the interval AB . | 1 |
| (ii) Find the gradient of the line AB . | 1 |
| (iii) What is the size of the acute angle between the line AB and the line l ? | 1 |
| (iv) Show that the equation of the line AB is $x + y - 2 = 0$. | 1 |
| (v) Copy the diagram into your writing booklet and shade the region defined by $x + y - 2 < 0$. | 1 |
| (vi) Write down the equation of the line l . | 1 |
| (vii) The point C is on the line l such that AC is perpendicular to AB . Find the coordinates of C . | 2 |
| (b) Let A be the point $(3, -1)$ and B be the point $(9, 2)$. | 2 |
- Find the coordinates of the point P which divides the interval AB externally in the ratio $5:2$.

QUESTION 7**10 marks**

Start a NEW answer booklet

Marks

(a) Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1}$.

2

(b) Given that $\frac{d}{dx}(\sin x) = \cos x$, find the derivative of $y = (x^2 - 5)\sin x$.

2

(c) Given that $f(x) = x^3 - 3x$ find the equation of the tangent to the curve $y = f(x)$ at the point on the curve where $x = -1$.

3

(d) Differentiate:

3

$$y = \frac{x^2}{(2x+1)^3}, x \neq -\frac{1}{2}$$

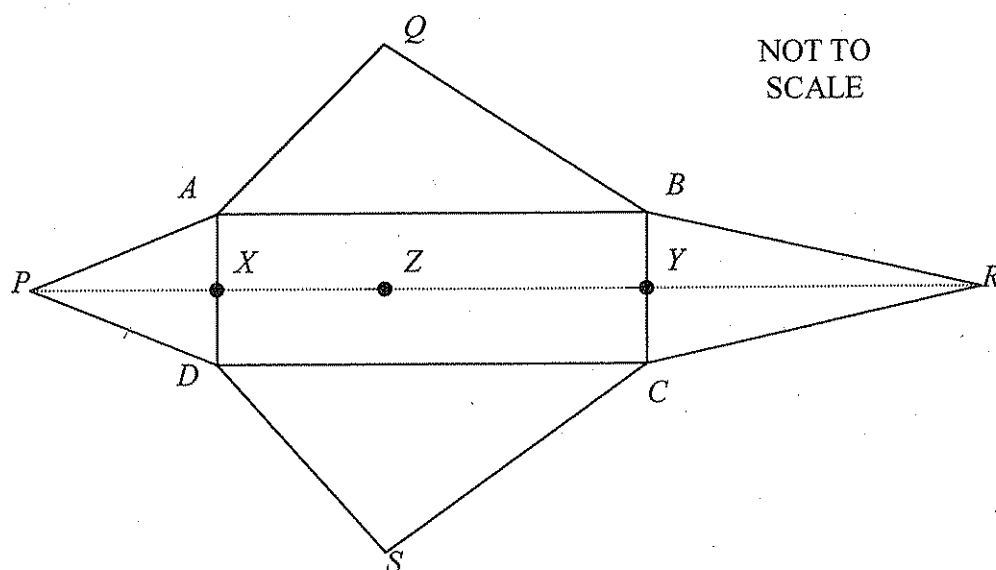
QUESTION 8

10 marks

Start a NEW answer booklet.

Marks

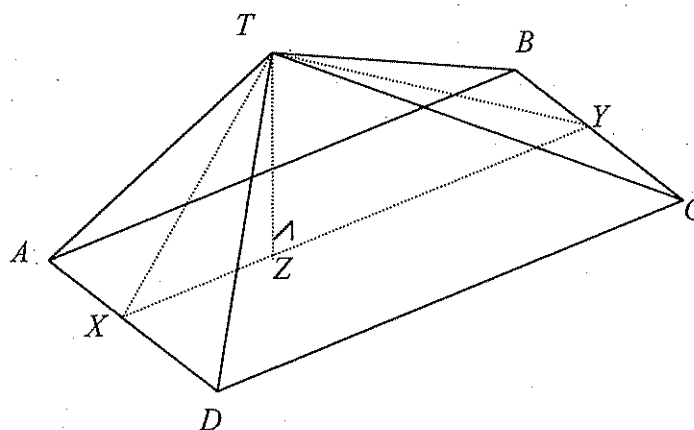
(a)



The figure above shows the net of an oblique pyramid with a rectangular base.

In this figure, $PXZYR$ is a straight line, $PX = 15$ cm, $RY = 20$ cm, $AB = 25$ cm and $BC = 10$ cm. Further, $AP = PD$ and $BR = RC$.

When the net is folded, points P , Q , R , and S all meet at the apex T , which lies vertically above the point Z in the horizontal base, as shown below.



- (i) Show that $\triangle TXY$ is right-angled. 1
- (ii) Hence, show that T is 12 cm above the base. 2
- (iii) Hence, find the angle that the face DCT makes with the base. 2

Question 8 continues on next page.

QUESTION 8 (continued)

Marks

- (b) Prove the identity:

2

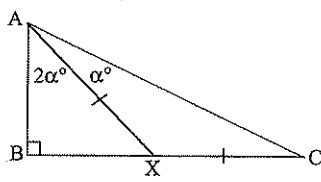
$$\operatorname{cosec} 2A + \cot 2A = \cot A.$$

- (c) Solve the equation below for $0^\circ \leq \theta \leq 360^\circ$:

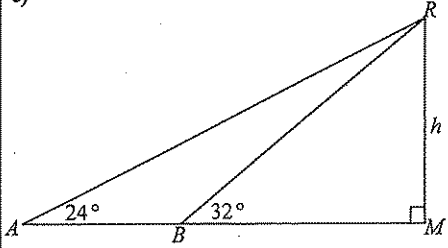
3

$$5\cos^2 \theta + 2\sin \theta - 2 = 0$$

Year 11 Mathematics Extension 1 Question No. 1 Solutions and Marking Guidelines		Yearly Examination 2011
Outcomes Addressed in this Question		
P3	performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities	
P4	chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques	
PE3	solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations	
Outcome	Solutions	Marking Guidelines
(a) P4	$\log_2 5 = \frac{\log 5}{\log 2}$ $= 2.321928095$ ≈ 2.3219	Award 2 for correct answer Award 1 attempts to use "change of base" and round to 4 d.p.
(b) P4	$3 \times 2^{2.5} = 16.97056275$ ≈ 17.0	Award 1 for correct answer
(c) P3	$\frac{1+\sqrt{3}}{2-\sqrt{3}} = \frac{1+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$ $= \frac{5+3\sqrt{3}}{4-3}$ $= 5+3\sqrt{3}$	Award 2 for correct solution Award 1 multiplies by the conjugate
(d) PE3	$\frac{3-2x}{x-3} < 4 \quad (x \neq 3)$ $(3-2x)(x-3) < 4(x-3)^2$ $4(x-3)^2 - (3-2x)(x-3) > 0$ $(x-3)(4(x-3) - (3-2x)) > 0$ $(x-3)(6x-15) > 0$ $\therefore x < 2\frac{1}{2} \text{ or } x > 3$	Award 3 for correct solution Award 2 multiplies both sides by $(x-3)^2$ and attempts to solve the resulting quadratic inequality. Award 1 multiplies both sides by $(x-3)$ and attempts to solve the resulting linear inequality.
(e) PE3	Consider $a + \frac{1}{a} - 2$ $a + \frac{1}{a} - 2 = \frac{a^2 + 1 - 2a}{a}$ $= \frac{(a-1)^2}{a}$ If $a > 0$, $(a-1)^2 \geq 0$ and $\frac{(a-1)^2}{a} \geq 0$ $\therefore a + \frac{1}{a} - 2 \geq 0$ $\therefore a + \frac{1}{a} \geq 2$	Award 2 for correct solution Award 1 for attempting to prove the result by a valid means

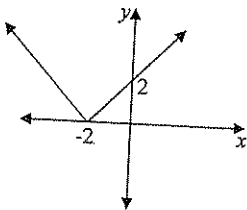
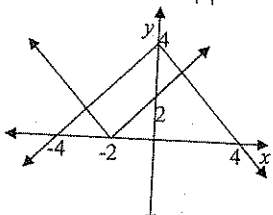
Year 11 Mathematics Extension 1 Yearly Examination 2011		
Question No. 2		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
P4 Chooses and applies appropriate geometric techniques.		
Outcome	Solutions	Marking Guidelines
P4	(a) $\angle BCE = 93^\circ + 61^\circ \text{ (Exterior angle of } \triangle ABC \text{ equals the sum of the two interior opposite angles.)}$ $\therefore \angle BCE = 154^\circ$ $\angle BCD = \frac{154^\circ}{2} \text{ (CD bisects } \angle BCE \text{)}$ $\therefore \angle BCD = 77^\circ$	2 marks Correct answer with full reasons. 1 mark Correct answer with incomplete reasons OR incorrect answer with relevant and correct reasoning.
P4	(b)(i) 	1 mark Correctly drawn diagram showing all given information
P4	(ii) $AX = XC \text{ (given)}$ $\therefore \triangle AXC \text{ is isosceles (two equal sides)}$ Now, $\angle ACX = \alpha^\circ$ (angles opposite equal sides in isosceles $\triangle AXC$) In $\triangle ABC$, $2\alpha + \alpha + \alpha + 90^\circ = 180^\circ \text{ (angle sum of triangle)}$ $4\alpha = 90^\circ$ $\therefore \alpha = 22.5^\circ$	2 marks Correct answer with full reasons. 1 mark Correct answer with incomplete reasons OR incorrect answer with relevant and correct reasoning.
P4	(c)(i) $BX \parallel CZ$ ($BXZC$ is a parallelogram) $AX \parallel CZ$ (BX produced to A) In $\triangle AXY$ and $\triangle CZY$ $\angle AXY = \angle CYZ \text{ (vertically opposite angles are equal)}$ $AY = CY \text{ (Y is the midpoint of AC)}$ $\angle XAY = \angle ZCY \text{ (alternate angles equal } AX \parallel CZ \text{)}$ $\therefore \triangle AXY \cong \triangle CZY \text{ (AAS)}$	3 marks Correct solution 2 marks Substantial progress towards correct solution 1 mark Limited progress towards correct solution
P4	ii) $AX = CZ \text{ (matching sides of congruent triangles are equal)}$ $BX = CZ \text{ (opposite sides are equal, } BXZC \text{ is a parallelogram)}$ $\therefore AX = BX$ $\therefore X \text{ is the midpoint of } AB.$	2 marks Correct solution 1 mark Substantial progress towards correct solution

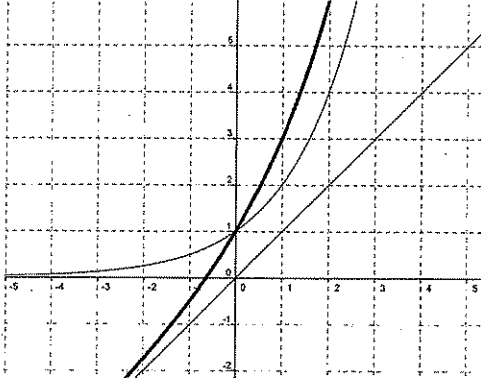
Year 11 Question No. 3	Extension 1 Mathematics Yearly Solutions and Marking Guidelines	Examination 2011
Outcomes Addressed in this Question		
PE2 - uses multi-step deductive reasoning in a variety of contexts P3 - performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities P4 - chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques P5 - understands the concept of a function and the relationship between a function and its graph PE5 - determines derivatives which require the application of more than one rule of differentiation P6 - relates the derivative of a function to the slope of its graph PE6 - makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations P8 - understands and uses the language and notation of calculus		
	Sample Solution	Marking Guidelines
P4	3. a) $LHS = \frac{2 \tan \frac{\pi}{6}}{1 - \tan^2 \frac{\pi}{6}}$ $= \tan 2\left(\frac{\pi}{6}\right)$ $= \tan \frac{\pi}{3}$ Since, double angle result for tan is: $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ $\text{Therefore, } \frac{2 \tan \frac{\pi}{6}}{1 - \tan^2 \frac{\pi}{6}} = \tan \frac{\pi}{3}$	2 marks for complete correct solution 1 marks for partially correct solution
P4	b) $\sin 75^\circ$ $= \sin(30^\circ + 45^\circ)$ $= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$ $= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$ $= \frac{1 + \sqrt{3}}{2\sqrt{2}}$	2 marks for complete correct solution 1 marks for partially correct solution

	c) 	
P4	i) $\frac{BR}{\sin 24^\circ} = \frac{1}{\sin 8^\circ}$ $BR = \frac{\sin 24^\circ}{\sin 8^\circ}$ $\cos 32^\circ = \frac{BM}{BR}$ $\therefore BM = BR \cos 32^\circ$ $= \frac{\sin 24^\circ}{\sin 8^\circ} \cos 32^\circ$ $= 2.478440398$ $= 2.478 \text{ km or } 2478 \text{ m}$	2 marks for complete correct solution 1 marks for partially correct solution
P4	ii) $\frac{h}{BM} = \tan 32^\circ$ $h = BM \tan 32^\circ$ $h = \left(\frac{\sin 24^\circ}{\sin 8^\circ} \cos 32^\circ \right) \tan 32^\circ$ $= 1.548701446$ $= 1.549 \text{ km or } 1549 \text{ m}$	1 mark for correct solution
P3	d) $LHS = \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta}$ $= \frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta \cos^2 \theta}$ $= \frac{1}{\sin^2 \theta \cos^2 \theta}$ $= \frac{1}{\sin^2 \theta} \times \frac{1}{\cos^2 \theta}$ $= \operatorname{cosec}^2 \theta \sec^2 \theta$ $= RHS$ $\therefore \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta$	3 marks for complete correct solution 2 marks for partially correct solution no reasoning 1 mark for any correct statement

Outcomes Addressed in this Question

- P4** Chooses and applies appropriate algebraic, trigonometric and geometric techniques
P5 Understands the concept of a function and the relationship between a function and its graph
PE3 Solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations.
PE6 Makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations.

Outcome	Solutions	Marking Guidelines
P4	(a)(i) A function is odd if $f(-x) = -f(x)$ $f(x) = 2x^3$ $f(-x) = 2(-x)^3$ $= -2x^3$ $= -(2x^3)$ $= -f(x)$	2 marks : correct definition and correct solution 1 mark : one of above
P5	(b)(i) Axis of symmetry of $y = ax^2 + bx + c$ is given by $x = \frac{-b}{2a}$ $= \frac{-4}{2}$ $\therefore$ axis of symmetry is $x = -2$ (ii) Vertex is on the axis of symmetry. Substituting $x = -2$ into $y = x^2 + 4x$, $y = 4 - 8 = -4$ $\therefore$ vertex is $(-2, -4)$ As this is a concave up parabola, range is $y \geq -4$.	1 mark : correct answer 2 marks : correct vertex & range 1 mark : one of above
P5	(c)(i) 	2 mark : correct graph 1 mark: incomplete graph
PE3	(ii) Sketching $y = 4 - x$ on the same diagram,  $x+2 > 4 - x$ when the graph of $y = x+2$ is above the graph of $y = 4 - x$.	2 marks : correctly graphs $y = 4 - x$ and correct solution to inequality 1 mark : substantial progress towards correct solution, by using the given graph

PE6	The graphs meet when $y = x + 2$ and $y = 4 - x$ intersect and when $y = -(x + 2)$ and $y = 4 - (-x)$ intersect. Solving $x + 2 = 4 - x$, $2x = 2$, $\therefore x = 1$. Solving $-x - 2 = 4 + x$, $2x = -6$, $\therefore x = -3$. From the graph, it can be seen that the solution to the inequality is given by $x < -3$ and $x > 1$. (d) Add the y values at particular values of x. 	1 mark : correct solution
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Outcomes Addressed in this Question

P4 chooses and applies appropriate arithmetic, algebraic, and trigonometric techniques

P7 determines the derivative of a function through routine application of the rules of differentiation

HE2 uses inductive reasoning in the construction of proofs

Outcome	Solutions	Marking Guidelines
P4	$(a) (i) 36^\circ = \frac{36\pi}{180}$ $= \frac{\pi}{5}$	1 mark: correct answer
P4	$(ii) A = \frac{1}{2}r^2(\theta - \sin \theta)$ $= \frac{1}{2} \times 8^2(1.7 - \sin 1.7)$ $= 22.7 \text{ cm}^2$	2 marks: correct solution 1 mark: substantially correct solution
P7	$(b) (i) \frac{d}{dx} \left(5x^4 - \frac{4}{x} \right) = 20x^3 + \frac{4}{x^2}$	2 marks: correct solution 1 mark: substantially correct solution
P7	$(ii) \frac{d}{dx} \sqrt{2x+7} = \frac{d}{dx} (2x+7)^{\frac{1}{2}}$ $= \frac{1}{2}(2x+7)^{-\frac{1}{2}} \times 2$ $= \frac{1}{\sqrt{2x+7}}$	2 marks: correct solution 1 mark: substantially correct solution
HE2	(c) Show true for $n = 1$ $\text{LHS} = \frac{1}{(3 \times 1 - 1)(3 \times 1 + 1)}$ $= \frac{1}{4}$ $\text{RHS} = \frac{1}{3(1) + 1}$ $= \frac{1}{4}$ Assume true for $n = k$. $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3k-2)(3k+1)} = \frac{k}{3k+1}$ Prove true for $n = k + 1$ $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3[k+1]-2)(3[k+1]+1)}$ $= \frac{k+1}{3(k+1)+1}$ i.e. $\frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} = \frac{k+1}{3k+4}$	3 marks: correct solution 2 mark: substantially correct solution 1 mark: partially correct solution

$$\begin{aligned}
 \text{LHS} &= \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} \\
 &= \frac{k(3k+4)}{(3k+1)(3k+4)} + \frac{1}{(3k+1)(3k+4)} \\
 &= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)} \\
 &= \frac{(3k+1)(k+1)}{(3k+1)(3k+4)} \\
 &= \frac{k+1}{3k+4} \\
 &= \text{RHS}
 \end{aligned}$$

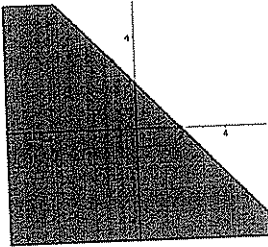
$\therefore$ Proven by mathematical induction for $n > 0$.

Year 11 Yearly 2011 – Extension 1 Mathematics

Question No: 6

Solutions and Marking Guidelines

Outcome Addressed in this Question: Chooses and applies appropriate coordinate techniques to solve problems.

	Sample Solution	Marking Guidelines
a)	i) $\sqrt{18}$	1 mark – correct solution
	ii) -1	1 mark – correct solution
	iii) 45°	1 mark – correct solution
	iv) $y - 0 = -1(x - 2)$ $y - 0 = -x + 2$ $\therefore x + y - 2 = 0$	1 mark – correct equation derived with all steps clearly shown
	v) 	1 marks – correct region shaded with dotted line clearly shown.
	vi) $x = 2$	1 mark – correct solution
	vii) $\text{gradient of AC} = 1$ $\therefore \text{Equation of AC is } y - 3 = 1(x + 1)$ $y - 3 = x + 1$ $y = x + 4$ $\text{When } x = 2, y = 2 + 4 = 6$ $\therefore C(2, 6)$	2 marks – correct solution 1 mark – substantial progress towards correct solution
b)	$(3, -1) \quad (9, 2)$ $5 : -2$ $P\left(\frac{5 \times 9 + -2 \times 3}{5 + (-2)}, \frac{5 \times 2 + -2 \times -1}{5 + (-2)}\right)$ $P\left(\frac{45 - 6}{3}, \frac{10 + 2}{3}\right)$ $P(13, 4)$	2 marks – correct solution 1 mark – substantial progress towards correct solution

Year 11 Yearly 2011 – Extension 1 Mathematics Solutions and Marking Guidelines Question No: 7 Outcome Addressed in this Question: Determines derivatives which require various rules of differentiation		
	Sample Solution	Marking Guidelines
a)	$\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1}$ $= \lim_{x \rightarrow 1} \frac{(x-1)(x-2)}{x-1}$ $= \lim_{x \rightarrow 1} (x-2)$ $= 1 - 2$ $= -1$	2 marks – correct solution 1 mark – substantial progress towards correct solution
b)	$y' = 2x \sin x + (x^2 - 5) \cos x$	2 marks – correct solution 1 mark – substantial progress towards correct solution
c)	$f'(x) = 3x^2 - 3$ at $x = -1$ $f'(x) = 0$ $f(x) = 2$ $\therefore$ Equation of tangent is $y = 2$	3 marks – correct solution 2 marks – correctly calculates $f(x)$ and $f'(x)$ 1 mark – correctly calculates $f(x)$ or $f'(x)$
d)	$y' = \frac{(2x+1)^3 \cdot 2x - x^3 \cdot 3(2x+1)^2 \cdot 2}{((2x+1)^3)^2}$ $= \frac{2x \cdot (2x+1)^3 - 6x^3 \cdot (2x+1)^2}{(2x+1)^6}$ $= \frac{2x \cdot (2x+1)^2 (2x+1 - 3x)}{(2x+1)^6}$ $= \frac{2x \cdot (1-x)}{(2x+1)^4}$	3 marks – correct solution 2 mark – correctly differentiates y , but does not simplify correctly 1 mark – substantial progress towards correct derivative

Year 11 Question No. 8	Mathematics Extension Solutions and Marking Guidelines	Task 3 2011
Outcomes Addressed in this Question		
P4 chooses and applies appropriate arithmetic, algebraic, and trigonometric techniques		
Outcome	Solutions	Marking Guidelines
	(a) (i) $TX^2 + TY^2 = PX^2 + RY^2$ $= 15^2 + 20^2$ $= 625$ $= XY^2$ $\therefore \triangle TXY$ is right angled. (ii) $A = \frac{1}{2} \times 15 \times 20$ $= 150$ $A = \frac{1}{2} \times XY \times TZ$ $150 = \frac{1}{2} \times 25 \times TZ$ $TZ = 12$ (iii) $\tan \theta = \frac{12}{5}$ $\theta = 67^\circ 23'$ (b) LHS = cosec $2A$ + cot $2A$ $= \frac{1}{\sin 2A} + \frac{\cos 2A}{\sin 2A}$ $= \frac{1 + \cos 2A}{\sin 2A}$ $= \frac{2 \cos^2 A}{2 \sin A \cos A}$ $= \frac{\cos A}{\sin A} = \cot A$ $= \text{RHS}$ (c) $5 \cos^2 \theta + 2 \sin \theta - 2 = 0$ $5(1 - \sin^2 \theta) + 2 \sin \theta - 2 = 0$ $5 \sin^2 \theta - 2 \sin \theta - 3 = 0$ $(5 \sin \theta + 3)(\sin \theta - 1) = 0$ $\sin \theta = -\frac{3}{5}$ or 1 $\theta = 216^\circ 52', 323^\circ 8', 90^\circ$	1 mark: correct solution 2 marks: correct solution 1 mark: substantially correct solution 2 marks: correct solution 1 mark: substantially correct solution 2 marks: correct solution 1 mark: substantially correct solution 3 marks: correct solution 2 mark: substantially correct solution 1 mark: partially correct solution